Task 8: RAG-Based AI Chatbot System Documentation

Objective

To develop a Retrieval-Augmented Generation (RAG) based AI chatbot that combines semantic search with a language model to generate accurate and context-aware responses from a given corpus.

Technologies & Libraries Used

- sentence-transformers – for embeddings
- FAISS / sklearn – for similarity search
- LangChain / custom pipeline – for RAG flow
- Python-dotenv – environment variables
- Hugging Face / LLM APIs – response generation
- OS & Pathlib – file handling

Tools

- VS Code
- Python
- Hugging Face Models

System Workflow

1. Load the text corpus (corpus.txt)
2. Split text into smaller chunks
3. Generate embeddings for each chunk
4. Store embeddings in vector database (FAISS)
5. Accept user query
6. Convert query into embedding
7. Perform similarity search
8. Retrieve top relevant chunks
9. Pass retrieved context to LLM
10. Generate final response

Key Features

- Context-aware responses
- Fast semantic retrieval
- Scalable architecture

- Improved chatbot intelligence
- Works with large datasets

Test Cases Implemented

- Basic query retrieval
- Context-based questions
- Multi-query comparison
- Edge case handling
- Large dataset performance test

Errors Faced

- Slow retrieval with large corpus
- Irrelevant chunk retrieval
- Token limit issues
- Embedding mismatch

Solutions Implemented

- Optimized chunk size
- Used efficient embedding models
- Improved similarity search
- Limited token usage
- Preprocessed text data

Output

```
(venv) PS D:\Daily Task\Task_8> python task8.py
🔄 Loading document...
✂️Splitting text...
🧠 Creating embeddings...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
 faster downloads.
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 55/55 [00:00<00:00, 7033.78it/s]
BertModel LOAD REPORT from: sentence-transformers/paraphrase-MiniLM-L3-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
✅ System Ready!

💬 Ask a question (or type 'exit'): hi

🤖 Answer:
 I cannot answer that question based on the provided context. The context is a technical document and does not contain
conversational information.

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💬 Ask a question (or type 'exit'): What is the Pulse-Core in NexusVortex OS?

🤖 Answer:
 The Pulse-Core is a synchronization layer utilized by the NexusVortex Operating System. This layer does not rely on a
central scheduler; instead, it uses a "Temporal-Oscillator" that pulses every 0.002 microseconds to ensure all 12 micro
-shards remain in a state of "Quantum-Coherence."

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💬 Ask a question (or type 'exit'): How many shards are present in the system?

🤖 Answer:
 Based on the context, there are at least 9 shards mentioned:
* Shard 04
* Shards 05-06 (2 shards)
* Shard 07
* Shard 08
* Shards 09-11 (3 shards)
* Shard 12

This sums up to 1 + 2 + 1 + 1 + 3 + 1 = 9 shards.

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💬 Ask a question (or type 'exit'): What is Data Bleed and how is it prevented?

🤖 Answer:
 Data Bleed is a condition where unsynchronized shards write conflicting data to the V-FAT32 file system.

It is prevented by the system entering "Stasis Mode," where the Pulse-Core throttles all non-essential I/O to 5% of max
imum capacity.

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💬 Ask a question (or type 'exit'): What are the minimum RAM requirements?

🤖 Answer:
 A minimum of 14GB of LPDDR5 RAM is required.

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💬 Ask a question (or type 'exit'): What does vortex --status do?

🤖 Answer:
 `vortex --status` generates a hexadecimal health report for all 12 Shards. (0x00 indicates Offline, 0xFF indicates Opt
imal).

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💬 Ask a question (or type 'exit'): What does Error XV-402 indicate?

🤖 Answer:
 Based on the context provided, there is no information about what Error XV-402 indicates. The context only mentions Er
ror Code: XV-909.
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💬 Ask a question (or type 'exit'): Why is Neural-Link replacing keyboards?

🤖 Answer:
 The context states that the upcoming "Singularity" update (Q4 2026) will remove all kernel-level drivers for physical
keyboards and mice, and the OS will transition to a 100% "Neural-Link" and "Vocal-Gesture" API interface. However, the
context does not provide a reason *why* Neural-Link is replacing keyboards.

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💬 Ask a question (or type 'exit'): exit
```

Conclusion

The Task 8 system enhances chatbot capabilities by integrating retrieval and generation, resulting in more intelligent and context-aware AI systems suitable for real-world applications.